

(Membrane Air Separation)

Pre-Lab Plan
Unit Operations Lab #3
Date: 10/1/2025

Group 2, Tuesday, 1 PM section

Team Members:

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1. Experimental Objective (5 pts)

The objective of the lab is to perform membrane air separation using hollow fiber modules and obtain the following data: operating pressure, Tube side flow rate, Tube side oxygen level, ambient temperature, Shell side flow rate, and Shell side oxygen level. The experiment is performed under various pressures and flow conditions using parallel valve configurations, stopping once oxygen and flowrate are constant, resulting in approximately 15 total runs.

2. Experimental

The Membrane Air Separation lab will involve pressurized air, containing mostly O₂ and N₂, that will be sent through two membrane modules (columns) containing long hollow fibers. These non-porous fibers will allow some O₂ and N₂ to permeate and eventually exit the tube into a surrounding cylindrical chamber. This will create two streams, permeate and non-permeate, whose composition will be measured. Throughout the experiment the flowrates through the system and the tube-side pressure of the system will be manipulated. End results will be analyzed and reported.

2.1 Apparatus (5pts)

The apparatus to run this lab is a membrane air separator, that is capable of running in two configurations. A gas mixture of oxygen and nitrogen is fed at a regulated rate (comp 1,4,6,7) into the system. In the system two membranes (comp 8.1 ,8.2) are present to separate the oxygen in the mixture. The membrane can be configured into series and parallel through 3-way valves (comp 3.1, 3.2, 3.3, 3.4). In the end an oxygen concentration reader and air flow reader are positioned to measure the amount of air and oxygen present from the separation. With this apparatus the goal is to see how well the separation works at different pressures and what pressure works best for series and parallel configurations.

Apparatus Components

EQUIPMENT	DESCRIPTION
1. Regulating Valve and Gauge	Limits gas flow to 200 psig
2. Oxygen Concentration Reader (2.1, 2.2)	Device connected to end of membrane to read the amount of oxygen present at the end of the separation cycle.

3. Four 3-way Valves (3.1, 3.2, 3.3, 3.4)	These four valves allow the system to be run in parallel or in series.
4. Needle Valve	Can be manipulated to control the flow rate of tube-side effluent.
5. Air Flow Reader (5.1, 5.2)	Measures the flow rate of gas traveling tube-side and shell-side.
6. High-Pressure Air Cylinder	Stores the air that will be used in the experiment
7. Pressure Transducer	Reads the amount of pressure going into the system.
8. Membrane (8.1, 8.2)	Where air septation is taking place, can have two running in a series or 2 at the same time for parallel.

Figure for Apparatus:

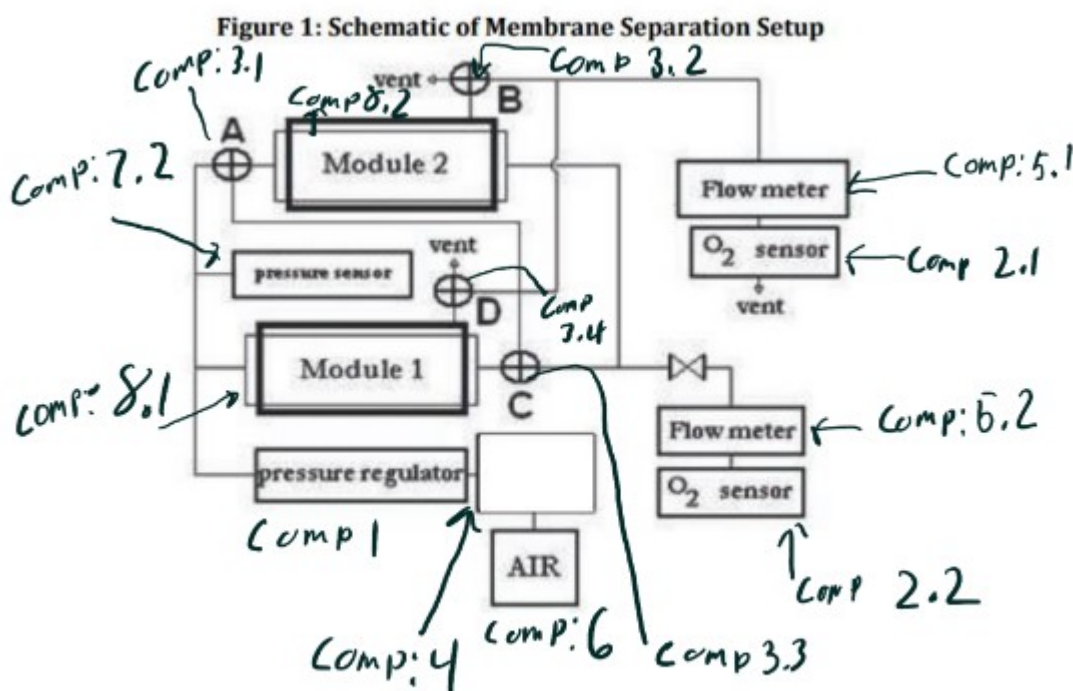


Fig 1

-Use of decimals for identifying due to the multiple components that are present more than once.

2.2 Materials and Supplies (5 pts)

MATERIALS & SUPPLIES	DESCRIPTION
1. Ultra Zero Compressed Air	This is pressurized purified air, that mainly consists of N ₂ and O ₂ , that will flow through the system and allow for a precise experiment with good data.
2. Power Strip	Supply power to equipment; also contains a ground fault circuit interrupter (GFCI) to protect against dangerous shocks. Also contains a 15-ampere overload protector.
3. Hollow Polymeric Fibers	These fibers act as non-porous semipermeable membranes that allow N ₂ and O ₂ to permeate and exit tube-side to enter the shell-side.

2.3 Experimental Plan (10 pts)

EXPERIMENTAL STEPS:

1. Begin by turning on oxygen analyzers and letting them reach a steady reading (~10 minutes). Manually calibrate if the percentage of O₂ indicated isn't around 20.9%.
2. Open the valve on the high-pressure air cylinder. Record the pressure inside of the cylinder.
3. Use regulator knob as needed to adjust pressure during the experiment.
4. Adjust 3-way valves for parallel operation: **Valve A** – left; **Valve B** – up to sample module 2 shell & down to vent (as needed); **Valve C** – down; and **Valve D** – up to sample module 1 shell & down to vent (as needed).
5. Do around 15 runs/trials that cover a range of operating pressures and tube-side flowrates. Each run/trial is finished when flowrates and O₂ levels are somewhat constant. (should take a few minutes)
6. Adjust 3-way valves for series operation: **Valve A** – right; **Valve B** - up to sample module 2 shell & down to vent (as needed); **Valve C** – up; **Valve D** - up to sample module 1 shell & down to vent (as needed).
7. Do around 15 runs/trials that meet the same requirements as for parallel operation.
8. Analyze and report data.

INDEPENDENT VARIABLES: Operating Pressure, Flowrates (tube-side & shell-side), Ambient Temperature (inside lab), Configuration (series or parallel), etc.

DEPENDENT VARIABLES: Composition, Percent O₂/N₂ Levels, Overall Mass Transfer Coefficients, Partial Pressures, Permeation Rates, Separation Coefficients, etc.

Project Safety Evaluation (10 pts)

	<i>Yes/No</i>	
Potential electrical hazards?	Yes	If yes, identify conditions: Although very unlikely, the power strip could surge during operation and cause an electrical fire.
Potential mechanical hazards?	No	If yes, identify conditions: No mechanical parts.
Potential pressure hazards?	Yes	If yes, identify conditions: A tank containing the air at high pressure is used to supply the membrane. Can explode if not secured tightly.
Potential temperature hazards?	No	If yes, identify conditions: No extreme temperatures present in lab.
Hazardous/toxic chemicals involved?	No	If yes, hazards are involved: No toxic/Hazardous chemicals involved, Gas is just Oxygen and Nitrogen and is dispersed out into the system.
Gloves required?	No	If yes, specify type: No extreme temperatures or exposure to dangerous chemicals present in lab,
MSDS reviewed	Yes	List MSDS sheets reviewed:

by all team members?		Oxygen: CAS Number 7782-44-7 Nitrogen: CAS number, 7727-37-9
<i>Process Parameters</i>	<i>Max Expected</i>	<i>List location and possible hazards and precautions</i>
Temperature (°C)	25°C	The process ran at room temperature.
Pressure (psia)	1000 psig 0 – 200 psig	At the tank holding the gas that is going to be processed through a membrane. Pressure of gas allowed to run in the membrane system.
<i>Safety Controls</i>	<i>yes/no</i>	<i>If yes, Provide description</i>
Dispersion Gauge	Yes	Limits the amount of gas flowing into membrane up to 200 psig.
Chains and Screws.	Yes	Chains surround the middle of the tank and are screwed to the table to steadily hold the tank in the middle.
Goggles	Yes	To prevent gas exposure into the eyes.

Reference:

Fig 1: McCabe W.L., Smith J.C., Harriot P., “Unit Operations in Chemical Engineering” 7th Edition pp. 882-922, (2005) \

Safety Statement

I have read relevant background material for the Unit Operations Laboratory entitled: “Membrane Air Separation” and understand the hazards associated with conducting this experiment. I have planned out my experimental work in accordance with standards and acceptable safety practices and will conduct all my experimental work in a careful and safe manner. I will also be aware of my surroundings, my group members, and other lab students, and will look out for their safety as well.

Team Member Printed Signature: Cesearo Dilosa

Date: 1 October 2025

Team Member Printed Signature: Joshua Laforest

Date: 1 October 2025

Team Member Printed Signature: Javier Silva

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